Question1

App.js

```
// Import functions from student.js
const {
    studentIntroduction,
    calculatePercentage
} = require('./student');
// Student information
const name = "Fatima Batool";
const course = "Web Development";
const obtainedMarks = 425;
const totalMarks = 500;
// Call functions
const introduction = studentIntroduction(name, course);
const percentage = calculatePercentage(
    obtainedMarks,
    totalMarks
);
// Display information
console.log("===== Student Information =====");
console.log(introduction);
console.log("Student Name:", name);
console.log("Course:", course);
console.log("Percentage:", percentage + "%");
console.log("Node.js Version:", process.version);
```

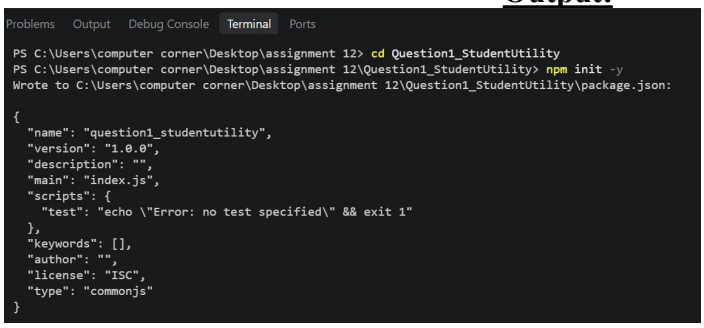
Student.js

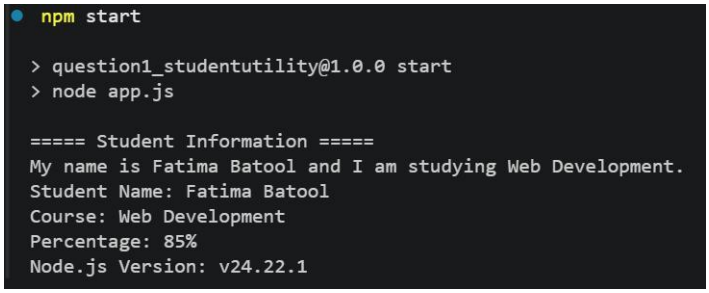
```
// Function to create student introduction
function studentIntroduction(name, course) {
    return `My name is ${name} and I am studying ${course}.`;
}
// Function to calculate percentage
function calculatePercentage(obtainedMarks, totalMarks) {
    return (obtainedMarks / totalMarks) * 100;
}
// Export both functions
module.exports = {
    studentIntroduction,
    calculatePercentage
};
```

Package.json

```
{
  "name": "question1_studentutility",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
 "scripts": {
    "start": "node app.js"
},
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
                                    }
```

Output:





Question 2

App.js

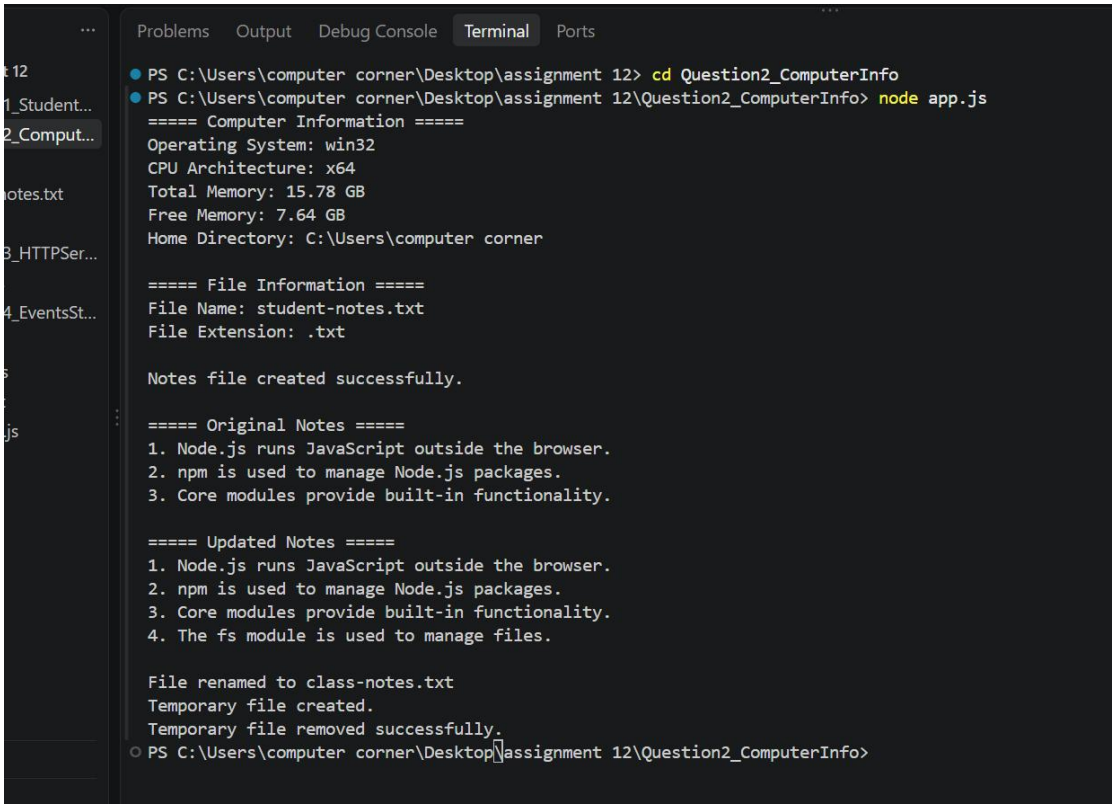
```
// Import core modules
const os = require('os');
const path = require('path');
const fs = require('fs');
// Display computer information
console.log("===== Computer Information =====");
console.log("Operating System:", os.platform());
console.log("CPU Architecture:", os.arch());
console.log(
    "Total Memory:",
    (os.totalmem() / (1024 ** 3)).toFixed(2),
    "GB"
);
console.log(
    "Free Memory:",
    (os.freemem() / (1024 ** 3)).toFixed(2),
    "GB"
);
console.log("Home Directory:", os.homedir());
// Create data folder if it does not exist
const folderPath = path.join(__dirname, 'data');
if (!fs.existsSync(folderPath)) {
    fs.mkdirSync(folderPath);
}
// Create file path using path module
const filePath = path.join(folderPath, 'student-notes.txt');
console.log("\n===== File Information =====");
console.log("File Name:", path.basename(filePath));
console.log("File Extension:", path.extname(filePath));
// Write notes into file
const notes = `1. Node.js runs JavaScript outside the browser.
2. npm is used to manage Node.js packages.
3. Core modules provide built-in functionality.`;
fs.writeFileSync(filePath, notes);
```

```
console.log("\nNotes file created successfully.");
// Read file contents
console.log("\n===== Original Notes =====");
console.log(fs.readFileSync(filePath, 'utf8'));
// Append another line
fs.appendFileSync(
    filePath,
    '\n4. The fs module is used to manage files.'
);
// Display updated contents
console.log("\n===== Updated Notes =====");
console.log(fs.readFileSync(filePath, 'utf8'));
// Rename file
const newFilePath = path.join(folderPath, 'class-notes.txt');
fs.renameSync(filePath, newFilePath);
console.log("\nFile renamed to class-notes.txt");
// Create temporary file
const tempPath = path.join(folderPath, 'temporary.txt');
fs.writeFileSync(tempPath, 'This is a temporary file.');
console.log("Temporary file created.");
// Remove temporary file after other operations
fs.unlinkSync(tempPath);
console.log("Temporary file removed successfully.");
```

Class-notes,txt

1. Node.js runs JavaScript outside the browser.
2. npm is used to manage Node.js packages.
3. Core modules provide built-in functionality.
4. The fs module is used to manage files.

Output:



Question 3

Server.js

```
// Import HTTP module
const http = require('http');
// Create HTTP server
const server = http.createServer((req, res) => {
    // Home route
    if (req.url === '/' && req.method === 'GET') {
        res.writeHead(200, {
            'Content-Type': 'text/plain'
        });
        res.end('Welcome to Our College!');
    }
    // About route
    else if (req.url === '/about' && req.method === 'GET') {
        res.writeHead(200, {
            'Content-Type': 'text/plain'
        });
        res.end(
            'Our college provides quality education and modern learning
facilities.'
        );
    }
    // Courses route
    else if (req.url === '/courses' && req.method === 'GET') {
        res.writeHead(200, {
            'Content-Type': 'text/html'
        });
        res.end(`
            <h1>Available Courses</h1>
            <ul>
                <li>Web Development</li>
                <li>Computer Science</li>
                <li>Information Technology</li>
            </ul>
        `);
    }
    // Students route
    else if (req.url === '/students' && req.method === 'GET') {
        const students = [
            { id: 1, name: 'Ali' },
            { id: 2, name: 'Ahmed' },
            { id: 3, name: 'Sara' }
        ];
        res.writeHead(200, {
            'Content-Type': 'application/json'
        });
        res.end(JSON.stringify(students, null, 2));
    }
    // Unknown route
    else {
        res.writeHead(404, {
            'Content-Type': 'text/plain'
        });
        res.end('Page Not Found');
    }
});
```

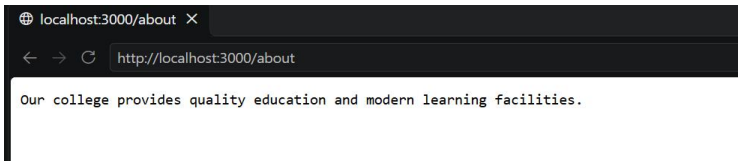
```
// Start server on port 3000
server.listen(3000, () => {
    console.log('Server is running at http://localhost:3000');
});
```

Output:

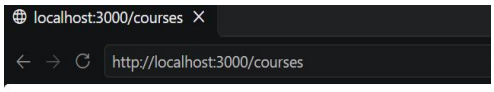
Home page:



About:



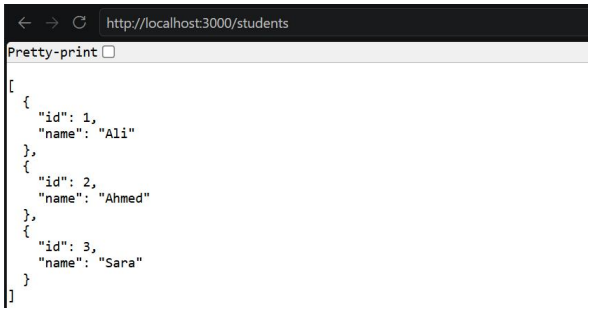
Courses



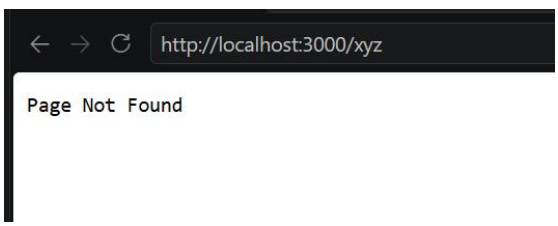
Available Courses

- Web Development
- Computer Science
- Information Technology

Students:



Unknown



Question 4

Events.js

```
// Import EventEmitter
const EventEmitter = require('events');
// Create EventEmitter object
const registrationSystem = new EventEmitter();
// Register event listener
registrationSystem.on('studentRegistered', (student) => {
    console.log(
        `Registration Successful! Student ID: ${student.id}, Name:
${student.name}`
    );
    // Schedule reminder after 3 seconds
    setTimeout(() => {
        console.log(
            `Reminder: Welcome ${student.name}! Your registration is
complete.`
        );
    }, 3000);
});
// Function to register a student
function registerStudent(id, name) {
    const student = {
        id: id,
        name: name
    };
    // Emit event with student details
    registrationSystem.emit('studentRegistered', student);
}
// Register two sample students
registerStudent(101, 'Ali');
registerStudent(102, 'Sara');
```

Streams.js

```
const fs = require('fs');
// Create large.txt
const logData = `
Student Ali registered successfully.
Student Sara logged into the system.
Student Ahmed downloaded course materials.
Student Ayesha submitted an assignment.
Student Usman completed a programming task.
Student Fatima logged out successfully.
`;
fs.writeFileSync('large.txt', logData);
console.log('large.txt created successfully.');
// Create readable stream
const readStream = fs.createReadStream('large.txt', {
    encoding: 'utf8',
    highWaterMark: 30
});
// Display each received chunk
readStream.on('data', (chunk) => {
    console.log('\nReceived Chunk:');
    console.log(chunk);
});
// Reading completed
readStream.on('end', () => {
    console.log('\nFile reading completed.');
});
```

```
// Reading error
readStream.on('error', (error) => {
    console.log('Reading Error:', error.message);
});
// Create streams for copying
const source = fs.createReadStream('large.txt');
const destination = fs.createWriteStream('copy.txt');
// Handle errors
source.on('error', (error) => {
    console.log('Source Error:', error.message);
});
destination.on('error', (error) => {
    console.log('Destination Error:', error.message);
});
// Copy file
source.pipe(destination);
// Copy completed
destination.on('finish', () => {
    console.log('\nFile copied successfully into copy.txt');
});
```

Copy.txt

Student Ali registered successfully.
Student Sara logged into the system.
Student Ahmed downloaded course materials.
Student Ayesha submitted an assignment.
Student Usman completed a programming task.
Student Fatima logged out successfully.

Output:

